

(19)



Europäisches Patentamt
European Patent Office
Office européen des brevets



(11) Publication number:

0 432 686 A1

(12)

EUROPEAN PATENT APPLICATION

(21) Application number: **90123684.4**

(51) Int. Cl.⁵ **D06P 1/607, D06P 3/32**

(22) Date of filing: **10.12.90**

(30) Priority: **11.12.89 JP 318932/89**

(43) Date of publication of application:
19.06.91 Bulletin 91/25

(84) Designated Contracting States:
AT BE CH DE DK ES FR GB GR IT LI LU NL SE

(71) Applicant: **NIHON SURFACTANT KOGYO K.K.**
24-3, Hasune 3-chome
Itabashi-ku Tokyo(JP)

(72) Inventor: **Ohba, Noriaki, c/o Nihon Surfactant Kogyo K.K.**
24-3, Hasune 3-chome
Itabashi-ku, Tokyo(JP)
Inventor: **Tabata, Yujin, c/o Nihon Surfactant Kogyo K.K.**
24-3, Hasune 3-chome
Itabashi-ku, Tokyo(JP)
Inventor: **Nagatomi, Tateyuki, c/o Nihon Surfactant Kogyo KK.**
24-3, Hasune 3-chome

Itabashi-ku, Tokyo(JP)
Inventor: **Nagatsuka, Masaaki, c/o Nihon Surfactant Kogyo KK.**
24-3, Hasune 3-chome
Itabashi-ku, Tokyo(JP)
Inventor: **Hashimoto, Satoru, c/o Nihon Surfactant Kogyo K.K.**
24-3, Hasune 3-chome
Itabashi-ku, Tokyo(JP)
Inventor: **Fujiu, Tomohisa, c/o Nihon Surfactant Kogyo K.K.**
24-3, Hasune 3-chome
Itabashi-ku, Tokyo(JP)
Inventor: **Klicker, Hellmut Hans-Georg**
5-29, Okuike-cho
Ashiya-shi, Hyogo-ken(JP)

(74) Representative: **Bühling, Gerhard, Dipl.-Chem. et al**
Patentanwaltsbüro Tiedtke-Bühling-Kinne
Grupe-Pellmann-Grams Bavariaring 4
W-8000 München 2(DE)

(54) **Leather dyeing auxiliary composition.**

(57) Dyeing auxiliary composition for leather comprises a polyoxyalkylene derivative of a condensation product of (a) one or more of polyalkylene (with 1 to 30 C) polyamines (n = 1 to 20), alkyl or alkenyl amines with 1 to 30 C, aromatic amines, and primary or secondary alkanolamines; (b) an aromatic compound having hydroxyl (OH) group, alkyl group with 1 to 10 C or alkenyl group with 1 to 10 C, and (c) formaldehyde or paraformaldehyde. With this composition, any anionic dyes may be used for surface dyeing of surface of grain and penetration dyeing to the inner layer of grain irrespective of tanning methods. Anionic surfactant promotes dyeing.

EP 0 432 686 A1

LEATHER DYEING AUXILIARY COMPOSITION

FIELD OF THE INVENTION

This invention relates to a dyeing auxiliary composition for leather. More particularly, it relates to a dyeing auxiliary composition for leather employed in dyeing a leather which may be dyed by any anionic dyes commonly employed in leather dyeing.

BACKGROUND

Prior Art

Heretofore, in leather dyeing, in order to cause the dye to be penetrated sufficiently into the inner layer of the grain and/or to dye the surface of the grain to a level high color value within a reasonable dyeing time, the following methods have been known in general:

- a) The leather is dyed in an alkaline dyeing bath (pH, 9) in the presence of ammoniac water as a processing agent;
- b) quaternary ammonium salts of long-chain alkyl are used as the processing agent;
- c) a condensate of fatty acids and triethanolamine is used as the processing agent;
- d) a polyoxyalkylene derivative of fatty acid amides or aliphatic amines is used as the processing agent; or
- e) a polyoxyalkylene derivative of fatty acids or aliphatic alcohols is used as the processing agent.

On the other hand, the following technology is described in the prior-art Japanese Patent Kokai Publications:

- f) The JP Patent Kokai Publication 55-93884/1980 discloses a surfactant consisting of a mixture composed of C9 to C24 aliphatic alcohol oxyethylated by 3 to 120 times and a C8 to C20 primary or secondary amine oxyethylated by 6 to 80 times. This surfactant is thought to be used as a leather dyeing auxiliary composition.
- g) The JP Patent Kokai Publication 61-89382/1986 discloses a method of employing a quaternary ammonium salt as a dyeing auxiliary composition for dyeing tanned leather, hide or fur with the use of basic dyes.
- h) The JP Patent Kokai Publication 62-263390/1987 discloses a leather dyeing auxiliary composition consisting of a mixture of (i) an alkylene oxide adduct to an aliphatic or aromatic compound containing one or more hydroxyl groups and (ii) a reaction product, having 170 to 20,000 molecular weight and carboxyl group, of an aliphatic, alicyclic or aromatic compounds containing at least two carboxyl groups and aliphatic or alicyclic compounds containing at least two hydrofyll groups and/or primary, secondary or tertiary amino groups etc.
- i) The JP Patent Kokai Publication 63-21987/1988 discloses a method of dyeing a leather with basic dyes and/or anionic dyes in which the anionic sulfo-group containing composition is used as the dyeing auxiliary composition.
- j) The JP Patent Kokai Publication 1-139882/1989 discloses a method of dyeing a leather using anionic metal complex dyes of specific structures, wherein a polymer with a cyanamide or a dicyandiamide of a mono-or polyfunctional amine containing one or more amino groups, or a reaction product thereof with an N-methylene derivative or epihalohydrine, is used as a dyeing auxiliary composition.

Problems in the Art

With the methods (a) to (e), penetration dyeing cannot be achieved sufficiently, even if a dye for penetration dyeing are used. That is, the dye can be hardly penetrated sufficiently into the inner layer of the grain within a practical dyeing time. Unless the dye is penetrated deep into the inner layer of the grain, flaws after dyeing will become apparent to degrade the product quality. The dyeing auxiliary compositions shown in (c) to (e) have insufficient operability because of excessive foaming.

On the other hand, with the techniques shown in (f) to (j) above, problems are first raised that penetration dyeing of the inner layer of the grain and surface dyeing of the grain cannot be achieved simultaneously, and there is much to be desired. In addition, the following problems are raised:

In order for the dye to be penetrated sufficiently into the inner layer of the grain with the use of the conventional dyeing auxiliary composition, specific dyes for penetration dyeing need to be selectively employed, so that there is a limitation to the usable dyes. That is, the dyes for surface dyeing are suited to

moderate and level dyeing of the leather surface, however, they are not suited to dyeing the inner layer of the grain within the interior of the leather, even if the conventional dyeing auxiliary composition is used simultaneously. On the other hand, the dye for penetration dyeing has acceptable properties of penetration dyeing into the inner layer of the grain, however, they are not suited to satisfactory level dyeing on the surface of the grain with a high color value, even if the conventional dyeing auxiliary composition is used simultaneously.

For example, anionic dyes for leather dyeing are classified into a class of dyes for surface dyeing with a greater covering force, such as, in general, direct dyes or metal complex dyes, and a class of dyes for penetration dyeing, such as, in general, acid dyes (especially of the penetrating type). For sufficient dye penetration into the inner layer of the grain, it has been necessary to use a dye for penetration dyeing. However, when the conventional dyeing auxiliary composition is used to this end, surface dyeing is inferior, while penetration dyeing is also not satisfactory.

In addition, with the conventional dyeing method, it is necessary to use different dyeing auxiliary compositions with different leather tanning methods, that is, depending on whether the leather is tanned cationically or anionically, such that the conventional dyeing auxiliary agents are not fully satisfactory in the dyeing effects as a function of the leather tanning methods employed. That is, there are cases where the leather is tanned cationically by chrome tanning or anionically by vegetable tanning. In general, the leather tanned anionically is dyed by using a cationic dye, whereas the leather tanned cationically is dyed by using anionic dyes. Although the anionic dyes are customary and available in many types, the leather tanned anionically can be dyed only unsatisfactorily even with the use of conventional dyeing auxiliary composition. Above all, the conventional dyeing auxiliary compositions are not satisfactory if both the penetration dyeing and surface dyeing are to be realized simultaneously.

There is also known a two-step dyeing method, according to which, for sufficient dye penetration into the inner layer of the grain with the use of the dye for surface dyeing, a pre-dyeing step is performed with the use of about two-thirds of the required amount of the dye and a secondary dyeing is then performed with the use of the remaining one-third of the dye, wherein the dye used in the pre-dyeing step is preferably in the form of powders for improving dye penetration. This method involves a complicated and laborious operation, while the effect of penetration dyeing also remains unsatisfactory.

SUMMARY OF THE DISCLOSURE

It is an object of the present invention to provide a dyeing auxiliary composition for leather which is free from the above mentioned drawbacks of the prior art. Means to Solve the Problems

According to the present invention, there is provided a dyeing composition for leather comprising, as an indispensable component, polyoxyalkylene derivative of a condensation product of

(a) one or more of polyalkylene (with 1 to 30 carbon atoms) polyamines with 1 to 20 nitrogen atoms, alkyl or alkenyl amines with 1 to 30 carbon atoms, aromatic amines, primary or secondary alkanolamines (referred to as ingredient (a));

(b) an aromatic compound having one or more of a hydroxyl (OH) group, an alkyl group with 1 to 10 carbon atoms or an alkenyl group with 1 to 10 carbon atoms (referred to as ingredient (b)), and

(c) formaldehyde or paraformaldehyde (referred to as ingredient (c)).

According to the present invention, there is also provided a dyeing auxiliary composition for leather comprising, as indispensable components, a polyoxyalkylene derivative of a condensation product of

(a) one or more of polyalkylene (with 1 to 30 carbon atoms) polyamines with 1 to 20 nitrogen atoms, alkyl or alkenyl amines with 1 to 30 carbon atoms, aromatic amines, primary or secondary alkanolamines;

(b) an aromatic compound having one or more of a hydroxyl (OH) group, an alkyl group with 1 to 10 carbon atoms or an alkenyl group with 1 to 10 carbon atoms, and

(c) formaldehyde or paraformaldehyde, and an anionic surfactant.

As the anionic surfactant, one or more of sulfonic acid, sulfonates, sulphates or salts thereof, is preferably employed.

Meritorious Effects of the Invention

Since the dyeing auxiliary composition for leather according to the present invention contains, as an indispensable component a polyoxyalkylene derivative of a condensation product of (a) one or more of polyalkylene (with 1 to 30 carbon atoms) polyamines ($n=1$ to 20), alkyl or alkenyl amines with 1 to 30 carbon atoms, aromatic amines, primary or secondary alkanolamines; (b) an aromatic compound having one

or more of a hydroxyl (OH) group, an alkyl group with 1 to 10 carbon atoms or an alkenyl group with 1 to carbon atoms, and (c) formaldehyde or paraformaldehyde, penetration dyeing to the inner layer of the grain and level high color value dyeing of all types of leather may be realized simultaneously by a simple method with use of a sole type of the dyeing auxiliary composition which is used in conjunction with any type of
 5 known anionic dyes for leather, such as acidic dyes, direct dyes or metal complex salt containing dyes, irrespective of the tanning methods employed.

On the other hand, if the dyeing auxiliary composition containing the polyoxyalkylene derivative of the condensation product and an anionic surfactant as indispensable components is employed in conjunction with any known types of anionic dyes for leather, such as acidic dyes, direct dyes or metal complex dyes,
 10 penetration dyeing to the inner layer of the leather grain and level high color value dyeing of the leather surface may be realized simultaneously by a simple method with use of a sole type of the dyeing auxiliary composition irrespective of the tanning methods employed.

The dyeing auxiliary composition for leather according to the present invention is free of foaming during treatment and thus exhibits superior operability.

15

BRIEF DESCRIPTION OF THE DRAWINGS

Fig. 1 is a cross-sectional view of a leather.

20 PREFERRED EMBODIMENTS

The proportion, by weight, of the ingredients (a), (b) and (c) of the condensation product is preferably as follows:

The ratio (a)/(b) is preferably 5/95 to 95/5 (more preferably 20/80 to 80/20). The ratio ((a)+(b))/(c) is
 25 preferably 99/1 to 70/30 (more preferably 95/5 to 80/20).

The dyeing auxiliary composition for leather according to the present invention contains, as an indispensable component, a polyoxyalkylene derivative of a condensation product of the above mentioned amine or amines, the above mentioned aromatic compound and formaldehyde or paraformaldehyde. If the composition additionally contains an anionic surfactant, a more intensive effect may be displayed by the
 30 synergistic effects of the two components.

Although the mixing proportion of these two ingredients may be arbitrary, it is preferred that the polyoxyalkylene derivative of the above mentioned condensation product account for 2.0 to 98.0 wt% and the anionic surfactant account for 98.0 to 2.0 wt%.

The above mentioned amine or amines, the above mentioned aromatic compound and formaldehyde or
 35 paraformaldehyde are condensed in a manner, known per se, to a condensate. The condensation degree n is in the range from 1 to 50 and preferably n=2 to 10. Ethylene- and/or propylene oxide may be added to the above condensation product by any known method of adduction with an addition mol number n equal to 1 to 100 and preferably 10 to 70.

The polyalkylene (with 1 to 30 carbon atoms) polyamines (with 1 to 20 nitrogen atoms) and alkyl- or
 40 alkenylamines with 1 to 30 carbon atoms may also include substituents, such as alkyl-, alkenyl- and/or 2-hydroxyalkyl groups. As the polyalkylene (C = 1 to 30) polyamines (n = 1 to 20) preferred are aliphatic amines and/or aliphatic polyamines, and derivatives thereof.

Aliphatic amines embrace ethylene diamine, propylene diamine and/or hexamethylene diamine etc., and their derivatives substituted with alkyl-, alkenyl- or 2-hydroxyalkyl groups. Aliphatic polyamines embrace
 45 diethylene triamine, triethylene tetraamine, tetraethylene pentaamine, pentaethylene hexamine etc., and their derivatives substituted with alkyl-, alkenyl-, or 2-hydroxyalkyl groups. As the alkyl- or alkenyl (C = 1 to 30) amines preferred are aliphatic amines and their derivatives such as hexyl amine, lauryl amine, myristyl amine, palmityl amine, cetyl amine, oleyl amine, stearyl amine etc, and their derivatives substituted with alkyl-, alkenyl- or 2-hydroxyalkyl groups.

Examples of the aromatic amines include benzylamine, phenylamine, naphthylamine, phenylene dia-
 50 mine, methylphenylene diamine and/or xylene diamine. These aromatic amines may also include N-monoalkyl substituents.

Examples of the primary or secondary alkanol amines include ethanol amine, diethanol amine, N-methyl-ethanol amine or N-methyl-diethanol amine.

Examples of the aromatic compounds having at least one of the OH-groups, alkyl groups with 1 to 10
 55 carbon atoms and alkenyl groups with 1 to 10 carbon atoms, may include phenolic substances, benzenes with alkyl or alkenyl (C = 1 to 10) substituents, and derivatives thereof.

Examples of phenolic substances include phenol, resorcin, catechol, p-hydroxyphenol, cresol, dimethyl

phenol, alkyl or alkenyl phenols with 7 to 10 carbon atoms in the alkyl or alkenyl moiety, phenyl phenol, mono-, bis- or tris benzyl phenol, and bisphenol derivatives.

Examples of benzenes having substituents such as alkyl or alkenyl with 1 to 10 carbon atoms are methyl-substituted benzenes, such as toluene, xylene or mesitylene.

5 Preferred anionic surfactants are sulfonic acids, salts thereof, sulphates, and salts thereof. These anionic surfactants exhibit superior synergistic effects when employed with polyoxyalkylene derivatives of the above mentioned condensation product.

10 Examples of the anionic surfactants, which are sulfonic acid or salts thereof, may include sulfonates of alkanes (α -olefins, hydroxyalkanes or halogenated alkanes with 1 to 15 carbon atoms), straight-chained or branched alkylbenzene sulfonates with 1 to 15 carbon atoms, alkyl naphthalene sulfonates, formalin condensation products of alkyl naphthalene sulfonates, and polyoxyalkylenealkyl phenol sulfonic acid. These products are preferably in the form of salts, such as alkali metal salts, alkali earth metal salts, ammonium salts or organic amine salts.

15 Examples of anionic surfactants, which are sulphates and salts thereof may include sulfurated castor oil, fatty acid alkyl ester sulphates, alkyl sulphates, polyoxyalkylene alkyl ether sulphates, fatty acid monoglyceride sulphates and polyoxyalkylene alkyl phenyl ether sulphates. These are preferably in the form of salts, such as salts of alkali metals, alkaline earth metals, ammonium salts or organic amine salts.

The dyeing auxiliary composition for leather according to the present invention may be employed for both cationically tanned leather and anionically tanned leather. Examples of the cationically tanned leather 20 may include leather tanned by an inorganic tanning agent. Examples of the anionically tanned leather may include leather tanned by a vegetable tanning agent or an anionic synthetic tanning agent.

Leather tanning is performed by a predetermined method in a tanning drum or the like device, followed by neutralization and dyeing. Treatment by the dyeing auxiliary composition for leather according to the present invention may be performed prior to or simultaneously with dyeing or fatliquoring. By fatliquoring is 25 meant charging the fat into a bath following dyeing for imparting softness to the leather. An amount of fat corresponding to 1 to 5 wt% of dry leather is charged in this manner.

Treatment by the dyeing auxiliary composition according to the present invention is performed in a manner, known per se, in a tanning drum or the like apparatus. The treatment temperature is usually 10 to 95 °C and preferably 30 to 60 °C, the treatment time duration is usually 5 to 120 minutes and preferably 10 30 to 60 minutes, the bath ratio, that is the ratio of the leather weight to liquid weight, is usually 1:1 to 1:30 and preferably 1:1 to 1:10 and the concentration of the composition in the net state is usually 0.01 to 20 percent and preferably 0.05 to 10% based on the dried leather weight.

EXAMPLES

35 The present invention will be explained with reference to several Examples and Comparative Examples. These Examples, however, are given only for the sake of illustration and not for limiting the scope of the invention. In the following, the percent (%) denotes weight percent, unless otherwise specified.

40 Example 1

60% of p-methylphenol and 33% of ethylene diamine, each by weight, were charged in a stainless autoclave and melted to a homogeneous mass at 80-85 °C. Caustic potassium was added and stirred at 80-85 °C for 30 minutes. Then 7% by weight of paraformaldehyde was gradually added into the stainless steel 45 autoclave under stirring, followed by temperature rise upto 120 ° to 130 °C and reaction for 60 minutes. After the reaction, water was removed under vacuum to produce a yellow-brown pasty condensation product of the three starting ingredients. With continued stirring, a predetermined amount of ethylene oxide was continuously charged into the reaction system for four hours at a temperature of 120 ° to 130 °C under a pressure of 5.0 kgf/cm²G, until 50 mols of ethylene oxide (EO) were added per one mol of the 50 aforementioned condensation product. The resulting adduct was cooled down to give a dyeing auxiliary composition A for leather. This composition A was a water-soluble brown paste.

Example 2

55 60% of p-methylphenol, 33% of hexylamine and 7% of paraformaldehyde were condensed in the same way as in Example 1 to produce a condensation product, to which 50 mols of ethylene oxide were added to give a dyeing auxiliary composition B for leather.

Example 3

50% of nonylphenol, 43% of hexamethylene diamine and 7% of paraformaldehyde were condensed in the same way as in Example 1 to produce a condensation product to which 50 moles of ethylene oxide were added to produce a dyeing auxiliary composition C for leather.

Example 4

60% of xylene, 33% of cetylamine and 7% of paraformaldehyde were condensed in the same way as in Example 1 to produce a condensation product to which 50 mols of ethylene oxide were added to produce a dyeing auxiliary composition D for leather.

Example 5

55% of phenol, 38% of monoethanol amine and 7% of paraformaldehyde were condensed in the same way as in Example 1 to produce a condensation product to which 50 mols of ethylene oxide were added to produce a dyeing auxiliary composition E for leather.

Examples 6 to 40

40% of one of the dyeing auxiliary compositions A to E according to Examples 1 to 5, 10% of one of the following seven anionic surfactants and 50% of water were mixed together to produce 35 dyeing auxiliary compositions.

Anionic Surfactants

- i) sulfurated castor oil sodium salt
- ii) lauryl sulphate sodium salt
- iii) 2-ethylhexylalcohol sulphate sodium salt
- iv) L-dodecyl benzene sodium sulfonate
- v) P-methyl benzene sodium sulfonate
- vi) methylnaphthalene sodium sulfonate
- vii) C10 hydroxyalkane sodium sulfonate

Comparative Example 1

A composition consisting of a mixture of 50% of oleic acid amide (EO)₄ and 50% of water was produced.

Comparative Example 2

A composition consisting of a mixture of 50% of oleylamine (EO)₄ and 50% of water was produced.

Comparative Example 3

A composition consisting of a mixture of 50% of ethylene diamine (EO)₃(PO)₁ (tetronic compound) and 50% of water was produced.

Comparative Example 4

A composition consisting of a mixture of 50% of tallow reduced alcohol (EO)₅ and 50% of water was produced.

Comparative Examples 5 to 19

40% of one of the following five ingredients A, 10% of one of the following three ingredients B and 50% of water were mixed to produce 15 different compositions:

ingredient A:

- i) oleic acid amide (EO)₄ $\emptyset$
- ii) tallow reduced alcohol (EO)₅ $\emptyset$
- iii) ethylene diamine (EO)₃ $\emptyset$ (PO)₁ $\emptyset$
- 5 iv) oleylamine (EO)₄ $\emptyset$
- v) tallow amine (EO)₅ $\emptyset$

ingredient B:

- i) lauryl sulphate sodium salt
- 10 ii) methylnaphthalene sulfonic acid sodium salt
- iii) sulfurated castor oil sodium salt

Evaluation

The compositions according to the above Examples 1 to 40 and the compositions of the comparative
 15 Examples 1 to 19 were used for treatment of the leather to be dyed, and evaluation was made of the dyeing properties of the compositions. Dyeing was conducted on three different Bovine leather, that is, chrome Bovine leather, vegetable tanned Bovine leather and chrome tanned and subsequently vegetable tanned Bovine leather. The results of evaluation are as shown in tables 1, 2 and 3, respectively. The thickness of the three Bovine leather was in the range of 2 to 3 mm.

20 It is seen from these Tables that the dyeing auxiliary compositions of the present invention are markedly superior to the compositions of the Comparative Examples in both the penetration dyeing properties of penetrating into the inner layer of the grain and surface dyeing properties with respect to any types of leather and any types of the anionic dyes for leather without regard to the type of the tanning methodes employed.

Dyes

The following five dyes manufactured by Bayer AG were employed.

- i) Aciderm Brown 4G (acidic dyes; for penetration dyeing)
- 30 ii) Baygenal Brown CR(dyes for surface dyeing)
- iii) Bayegenal Brown LG (metal complex salt containing dyes with strong surface dyeing properties)
- iv) Bayenal Brown IT-V (dyes for penetration dyeing)
- v) Benzo Leather Fast Black VE (direct dyes with strong surface dyeing properties)

The methods of pre-treatment, penetration-dyeing-fatliquoring and evaluation for various types of leather
 35 are as follows:

Evaluation of Dyeing Properties of Chrome Tanned Bovine leather.

Method of Pre-treatment

40 The chrome Bovine leather tanned by the above described method was neutralized in a bath with a bath ratio of 1:2 with 2.0% of sodium formate and 2.0% of hydrogen sodium carbonate at 50 °C for 60 minutes. Meanwhile, percent (%) denotes the percent by weight based on the weight of the Bovine leather; hereinafter the same).

The Bovine leather was then washed with hot water of 50 °C for 30 minutes at a bath ratio of 1:5 and
 45 with flowing water so as to be used as samples.

Penetration-Dyeing-Fatliquoring Method

The Bovine leather following neutralization and washing was subjected to a penetration treatment for 30
 50 minutes at 50 °C at the bath ratio of 1:2.

In this case, 2.0% of the compositions of the Examples 1 to 5 and 4.0% of the compositions of the Examples 6 to 40 and Comparative Examples 1 to 19 were employed.

Then, without changing the bath, 2.0% of the dye was added to the same bath, and the dyeing was continued at 50 °C for 45 minutes.

55 After the termination of dyeing, 8.0% (on weight leather) of Chromopol UFB/w, manufactured by Stockhausen, Co, was added as the fatliquoring agent to the bath, which remained unchanged, and fatliquoring was continued at 50 °C for 30 minutes. After termination of fatliquoring, the bath was acidified with formic acid.

The leather treated in the above described manner was washed with water and dried.

Method of Evaluation

- 5 For evaluating the degree of dye penetration into the inner layer of the grain of the Bovine leather, pieces of about 1 cm in size were cut off at four corners of the dyed and dried leather for visually observing and evaluating the sectional surfaces.

The degree of surface dyeing was observed and checked by checking the state of level dyeing and thickening of the color hue.

- 10 Evaluation of Dyeing Properties of Vegetable Tanned Bovine leather:

The pre-treatment method and the method of evaluation after dyeing were conducted in the same way as in the evaluation of the dyeing properties on the chrome tanned Bovine leather.

Dyeing-Fatliquoring Method

15

After neutralization and washing, the leather was treated at a bath ratio of 1:2 at 50 °C for 30 minutes.

In this case, 2.0% of the compositions of the Examples 1 to 5 or 4.0% of the compositions of the Examples 6 to 40 or the Comparative Examples 1 to 19 were used as the dyeing auxiliary compositions.

- 20 Then, without changing the bath, 2.0% of the dye was added to the bath and dyeing was performed at 50 °C for 45 minutes.

After dyeing, 8.0% (on weight leather) of Chromopol UFB/w, manufactured by Stockhausen Co., were added to the bath, which had remained intact, to perform fatliquoring at 50 °C for 30 minutes. After termination of fatliquoring, the bath was acidified with formic acid.

The leather treated in the above manner was washed with water and dried.

- 25 Chrome Tanned and Subsequently Vegetable Tanned Bovine leather:

The pre-treatment, dyeing and fatliquoring method as well as the method of evaluation after dyeing, were performed in the same way as for evaluating the dyeing properties of the vegetable tanned Bovine leather.

- 30 Dyeing auxiliary composition for leather comprises a polyoxyalkylene derivative of a condensation product of (a) one or more of polyalkylene (with 1 to 30 C) polyamines (n = 1 to 20), alkyl or alkenyl amines with 1 to 30 C, aromatic amines, and primary or secondary alkanolamines; (b) an aromatic compound having hydroxyl (OH) group, alkyl group with 1 to 10 C or alkenyl group with 1 to 10 C, and (c) formaldehyde or paraformaldehyde. With this composition, any anionic dyes may be used for surface dyeing of surface of grain and penetration dyeing to the inner layer of grain irrespective of tanning methods.
- 35 Anionic surfactant promotes dyeing.

40

45

50

55

Table 1

Dyed Leather: chrome tanned Bovine leather

Dyes (mfd. by Bayer AG)											
		Aciderm Brown 4G		Baygenal Brown CR		Baygenal Brown LG		Baygenal Brown IT-U		Benzo Leather Fast Black VE	
		Dyeing Properties to Inner Layer of Grain	Surface Dyeing Properties	Dyeing Properties to Inner Layer of Grain	Surface Dyeing Properties	Dyeing Properties to Inner Layer of Grain	Surface Dyeing Properties	Dyeing Properties to Inner Layer of Grain	Surface Dyeing Properties	Dyeing Properties to Inner Layer of Grain	Surface Dyeing Properties
Ex.	1	●	●	●	○	●	○	●	●	●	○
	2	●	●	●	●	●	●	●	●	●	○
	3	●	○	●	○	●	○	●	○	●	●
	4	●	○	●	●	●	●	●	●	●	●
	5	●	○	●	○	●	○	●	○	●	○
	6-40	●	●	●	●	●	●	●	●	●	●
Comp. Ex.	1	Δ	x	x	x	x	x	Δ	x	x	x
	2	Δ	○	x	○	x	○	Δ	x	x	○
	3	Δ	○	x	○	x	○	Δ	x	x	○
	4	x	○	x	○	x	x	Δ	x	x	○
	5-19	Δ	○	x	○	x	○	Δ	x	x	○

Dyeing Properties to Inner Layer of Leather Grain

●...dyed to high color value
 Δ...only slightly dyed
 x...not dyed

Surface Dyeing Properties
 ●...dyed to high color value
 ○...moderately dyed
 x...only slightly dyed

Table 2

Dyed Leather: Vegetable tanned Bovine leather

		Dyes (mfd. by Bayer AG)			
		Aciderm Brown 4G		Baygenal Brown CR	
		Dyeing Properties to Inner Layer of Grain	Surface Dyeing Properties	Dyeing Properties to Inner Layer of Grain	Surface Dyeing Properties
Ex.	1	Δ	○	Δ	●
	2	Δ	○	Δ	●
	3	Δ	●	Δ	●
	4	Δ	○	Δ	●
	5	Δ	○	Δ	●
	6-40	Δ	●	Δ	●
Comp. Ex.	1	Δ	○	x	x
	2	x	x	x	○
	3	x	x	x	○
	4	Δ	○	x	x
	5-19	x	x	x	○

Dyeing Properties to Inner Layer of Grain

●...dyed to high color value

Δ...only slightly dyed

x...not dyed

Surface Dyeing Properties

●...dyed to high color value

○...moderately dyed

x...only slightly dyed

Table 3

Dyed Leather: Chrome tanned and subsequently vegetable tanned Bovine leather

		Dyes (mfd. by Bayer AG)			
		Aciderm Brown 4G		Baygenal Brown CR	
		Dyeing Properties to Inner Layer of Grain	Surface Dyeing Properties	Dyeing Properties to Inner Layer of Grain	Surface Dyeing Properties
Ex.	1	Δ	○	Δ	○
	2	Δ	○	Δ	○
	3	Δ	○	Δ	○
	4	Δ	○	Δ	○
	5	Δ	○	Δ	○
	6-40	Δ	⊙	Δ	⊙
Comp. Ex.	1	x	○	x	x
	2	x	x	x	○
	3	Δ	x	x	x
	4	x	○	x	x
	5-19	x	x	x	x

Dyeing Properties to Inner Layer of Grain

⊙...dyed to high color value

Δ...only slightly dyed

x...not dyed

Surface Dyeing Properties

⊙...dyed to high color value

○...moderately dyed

x...only slightly dyed

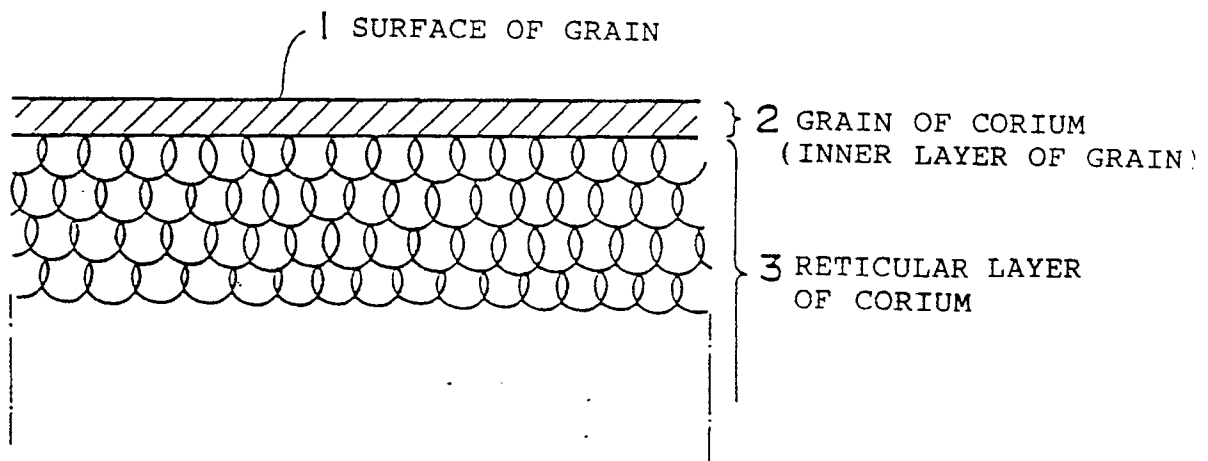
Claims

5

1. A dyeing auxiliary composition for leather comprising, as an indispensable component, a polyoxyalkylene derivative of a condensation product of
 - (a) one or more selected from the group of polyalkylene (with 1 to 30 carbon atoms) polyamines (n=1 to 20), alkyl or alkenyl amines with 1 to 30 carbon atoms, aromatic amines, and primary or secondary alkanolamines;
 - (b) an aromatic compound having one or more of a hydroxyl (OH) group, an alkyl group with 1 to 10 carbon atoms or an alkenyl group with 1 to 10 carbon atoms, and
 - (c) formaldehyde or paraformaldehyde.
2. A dyeing auxiliary composition for leather comprising, as indispensable components, a polyoxyalkylene derivative of a condensation product of
 - (a) one or more selected from the group of polyalkylene (with 1 to 30 carbon atoms) polyamines (n=1 to 20), alkyl or alkenyl amines with 1 to 30 carbon atoms, aromatic amines, and primary or secondary alkanolamines;
 - (b) an aromatic compound having one or more of a hydroxyl (OH) group, an alkyl group with 1 to 10 carbon atoms or an alkenyl group with 1 to 10 carbon atoms, and
 - (c) formaldehyde or paraformaldehyde, and an anionic surfactant.
3. The composition as defined in claim 2, characterized in that the anionic surfactant is one or more selected from the group consisting of sulfonic acid, sulfonates, sulphates, or salts thereof.
4. The composition as defined in claim 1, wherein the condensation degree n is in the range from 1 to 50 (preferably 2 to 10).
5. The composition as defined in any one of claims 1-4, characterized in that ethylene- and/or propylene oxide is added to a condensation mol number of 1 to 100 (preferably 10 to 70).
6. The composition as defined in any one of claims 1-5, characterized in that the polyalkylene polyamines and alkyl- or alkenylamines comprise substituents of alkyl-alkenyl-and/or 2-hydroxyalkyl groups.
7. The composition as defined in any one of claims 1-6, characterized in that the aromatic amines are selected from the group of benzylamine, phenylamine, naphthylamine, phenylene diamine, methyl-phenylene diamine and xylene diamine.
8. The composition as defined in any one of claims 1-7, characterized in that the primary or secondary alkanol amines are selected from the group consisting of ethanol amine, diethanol amine, N-methyl-ethanol amine and N-methyl-diethanol amine.
9. The composition as defined in any one of claims 1-8, characterized in that the aromatic compound is selected from the group of phenolic substances, benzenes with alkyl or alkenyl (with 1 to 10 carbon atoms) substituents, and derivatives thereof.
10. The composition as defined in claim 9, characterized in that the phenolic substances are selected from the group of phenol resorcin, catechol, p-hydroxyphenol, cresol, dimethyl phenol, alkyl or alkenyl phenols with 7 to 10 carbon atoms in the alkyl or alkenyl moiety, phenyl phenol, mono-, bis- or tris benzyl phenol, and bisphenol derivatives.
11. The composition as defined in any one of claims 3-10, characterized in that the sulfonic acid or salts thereof is/are selected from the group of sulfonates of alkanes, straight-chained or branched alkylbenzene sulfonates with 1 to 15 carbon atoms, alkylnaphthalene sulfonates, formalin condensation products of alkylnaphthalene sulfonates, and polyoxyalkylenealkyl phenol sulfonic acid,

12. The composition as defined in any one of claims 3-10, characterized in that the sulphates and salts thereof are selected from sulfurated castor oil, fatty acid alkyl ester sulphates, alkyl sulphates, polyoxyalkylene alkyl ether sulphates, fatty acid monoglyceride sulphates and polyoxyalkylene alkyl phenyl ether sulphates.
13. The composition as defined in any one of claims 2-12, characterized in that the polyoxyalkylene derivative of the condensation product amounts to 2.0 to 98.0 wt% and the anionic surfactant amounts to 98.0 to 2.0 wt%.
14. The composition as defined in claim 6, characterized in that the polyalkylene polyamines are selected from the group of aliphatic amines and aliphatic polyamines.
15. The composition as defined in claim 14, characterized in that the aliphatic amines are selected from the group of ethylene diamine, propylene diamine, hexamethylene diamine, and their derivatives substituted with alkyl-, alkenyl- or 2-hydroxyalkyl groups.
16. The composition as defined in claim 14, characterized in that the aliphatic polyamines are selected from the group of diethylene triamine, triethylene tetraamine, tetraethylene pentaamine, pentaethylene hexamine , and their derivatives substituted with alkyl-, alkenyl-, or 2-hydroxyalkyl groups
17. The composition as defined in claim 8, characterized in that alkyl- or alkenylamines are selected from the group of hexyl amine, lauryl amine, myristyl amine, palmityl amine, cetyl amine, oleyl amine, stearyl amine , and their derivatives substituted with alkyl-, alkenyl- or 2-hydroxyalkyl groups.

Fig. 1





European
Patent Office

EUROPEAN SEARCH REPORT

Application Number

EP 90 12 3684

DOCUMENTS CONSIDERED TO BE RELEVANT			
Category	Citation of document with indication, where appropriate, of relevant passages	Relevant to claim	CLASSIFICATION OF THE APPLICATION (Int. Cl.5)
A	EP-A-0 057 398 (BAYER AG) * the whole document * - - -	1-3,6, 8-12,15	D 06 P 1 607 D 06 P 3.32
A	FR-A-1 576 874 (BAYER AG) * the whole document * - - -	1-3,7,9, 10,14-17	
A	DATABASE WPI, no 70-15204R Derwent Publications Ltd, London, GB & CA-835829 (DIAMOND SHAMROCK CO.)(03.03.1970) * the whole abstract * - - - - -		
The present search report has been drawn up for all claims			TECHNICAL FIELDS SEARCHED (Int. Cl.5) D 06 P
Place of search The Hague		Date of completion of search 05 March 91	Examiner DELZANT J-F.
CATEGORY OF CITED DOCUMENTS E: earlier patent document, but published on, or after the filing date D: document cited in the application L: document cited for other reasons &: member of the same patent family, corresponding document X: particularly relevant if taken alone Y: particularly relevant if combined with another document of the same category A: technological background O: non-written disclosure P: intermediate document T: theory or principle underlying the invention			